CO-3

26/08/26

Data Storage System:

* This helps with the interaction between data creation & data flow, data management & data retrieval.
* When the volume and the formats of the data increase, sound storage systems help to preserve & protect the data's quality & availability.

- KDD ⇒ Knowledge Discovery Databases.

* Database servers are used to improve DBMS.

Ex:- SQL, Postgre & Mysql servers.

* Data Storage Systems support Data Warehouse, Big Data & File system.

* Hive ⇒ Hadoop Environment

Commonly used Technologies

1) Relational DBMS ⇒ Stores data in the form of tables with well defined formats.

ACID ⇒ Atomicity, Consistency, Isolation, Properties Durability.

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2) NoSQL Databases

* These databases are flexible when it comes to the models of data & are oriented at unstructured or semi-structured data.

Ex:- MongoDB, Cassandra, Redis, Neo4j etc..

3) Data Warehouses

* These are database types designed for processing & analytics of the data.

4) Data Lakes

* These storage repositories contain large volume of raw data in its most basic form for as long as it is will be required.

Ex:- Hadoop HDFS, S3 & Azure Data Lake storage.

5) Distributed File systems

* These systems have the ability of managing the data from one or more nodes.

* These machines are primarily meant for storage of massive data & computing at the same time.

Ex:- HDFS, GFS(Google File system).

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*In Big Data Analytics, the replication factor is 3, means the data is duplicated 3 times..

6. Object Storage

* This storage architecture or organization entails the storage of data as objects.

Eg:- Amazon S3, Azure Blob storage (Binary large objects)

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Data Engineering Storage Abstraction

* These abstractions are data organizations & query patterns that sit at the heart of the data engineering lifecycle.

* Storage abstraction is a conceptual layer that hides complexity of physical storage

File storage => HDFS, S3, GCS

Block " => EBS, Azure DLake

Object " => S3, MinIO

Table " => RDBMS, BigQuery.

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Key Abstractions

→ Data Lakes ⇒ Store raw structured/un-structured data.

→ Data Warehouses ⇒ Optimized for analytics.

→ Lakehouse ⇒ Hybrid model (Databricks Delta Lake, Apache Iceberg)

→ Streaming Storage ⇒ Real time event storage (Kafka, Pulsar).

Storage Layers

→ Curated Layer (Gold) ⇒ Analytics ready datasets.

→ Cleansed Layer (Silver) ⇒ Validated & enriched.

→ Raw Layer (Bronze) ⇒ Raw ingestion.

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Apache Kafka

* This is an open-source distributed streaming platform designed for real-time data processing and analysis.

Three Functions

1) Log Aggregation :-
   * Kafka acts as a central hub for collecting logs from multiple systems and applications.
   * These logs can then be analyzed to detect patterns.
2) Event streaming :- Kafka enable the continuous flow of data in real-time.
3) Data Integration :- Kafka connects different data sources & destinations, serving as a bridge between databases, microservices & analytics platforms.

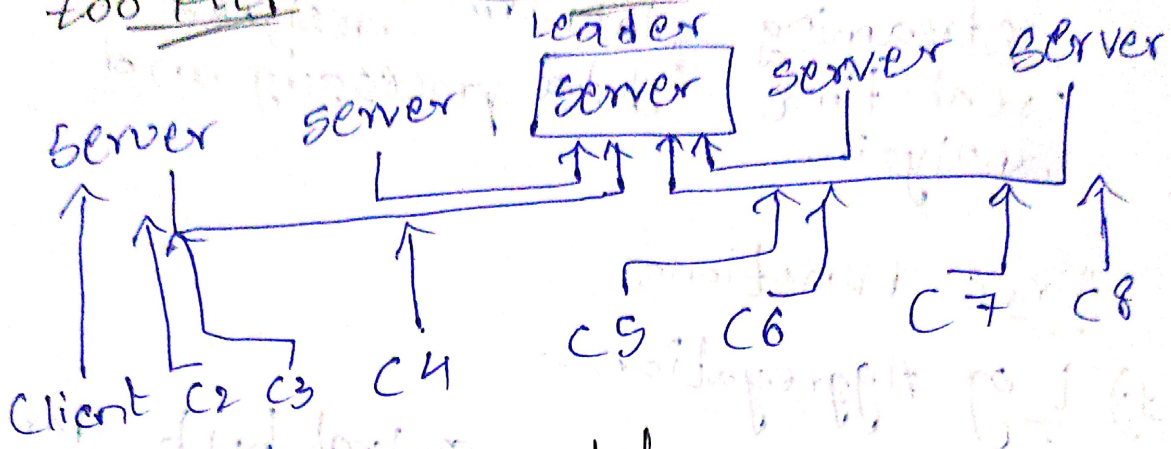
Apache Zookeeper

* Zookeeper is an open-source distributed coordination service that helps nodes in a distributed system communicate & stay synchronized.

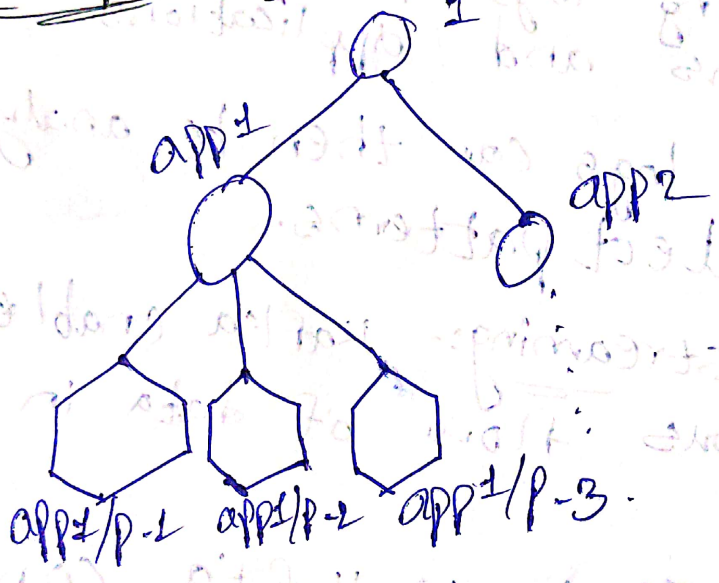
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* Maintains a hierarchial tree of data nodes called Znodes.

Zookeeper Architecture



Zookeeper Data Model



* Zookeeper provides APIs to create, read, update & delete Znodes.

Orchestration Frameworks

Apache Airflow

* Orchestration of complex workflows
Ex:- DAGs => Directed Acyclic Graphs.

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Dagster

* Next-gen orchestration with focus on data assets and observability.

Data Processing Technologies

Category	Tools
SQL	PostgreSQL, Redshift
Python Tools	pandas, NumPy, Pyspark
Orchestration	Airflow, Dagster
Transformation	dbt, Spark SQL

Apache Spark

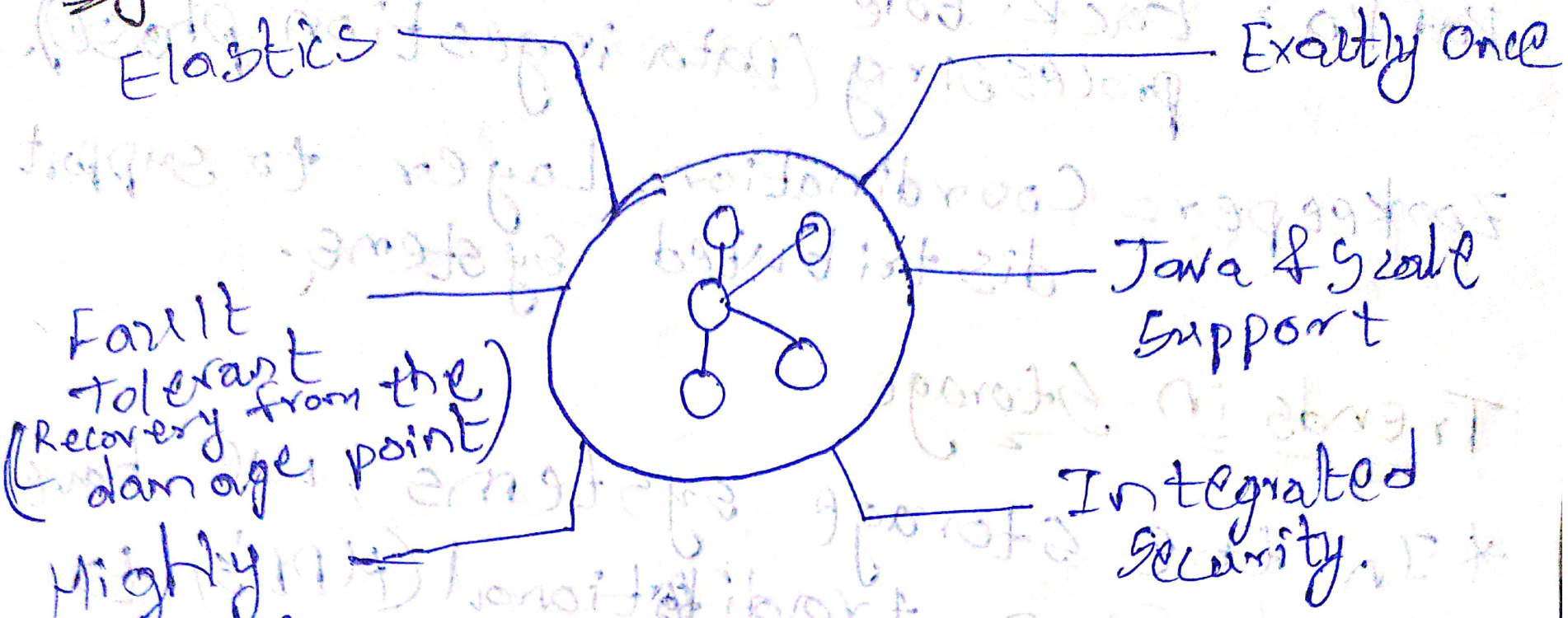
* Apache spark is an open-source, unified analytics engine for large-scale data processing.

Architecture

→ Driver Program
→ Cluster Manager
→ Executors
→ DAG Scheduler
→ Task scheduler.

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Key features of Kafka Streams



Single Tenant:- This storage isolates each tenant's data, storing each customer data in their own database.

Multi Tenant:- This allows multiple tenants to reside in the same database.

Apache Kafka = Data processing tool which satisfies data ingestion phase

Kafka + Zookeeper Integration

→ Zookeeper manages broker metadata, Quorum agreement

Advantages

→ Scalable
→ resilient
→ Low latency

Challenges

→ Operational Complexity
→ Requires careful tuning

Kafka = back-bone of real time data processing (Data ingestion phase).

Zookeeper = Coordination Layer to support distributed systems.

Trends in Storage

* In the storage systems, we have moved from traditional (HDD's) to SSD's, NVMe, Cloud storage, Software-defined storage, intelligent storages multi-cloud environments.

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Eight Major Trends

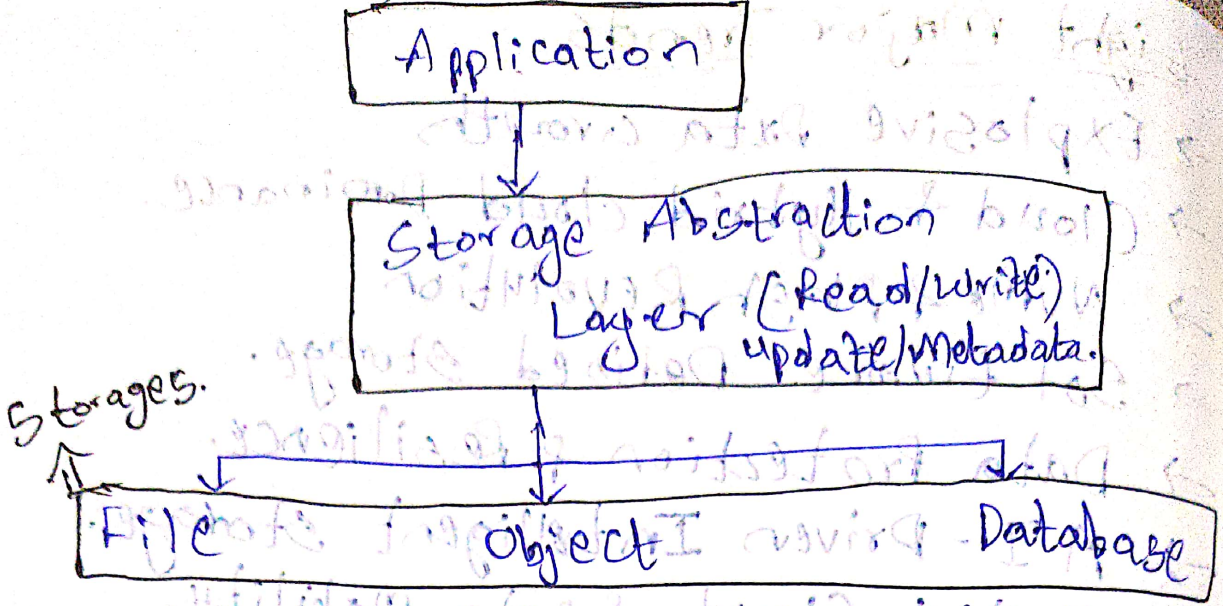
→ Explosive Data growth
→ Cloud & Hybrid cloud Dominance.
→ NVMe & Flash Revolution
→ Software Defined storage.
→ Data Protection & Resilience.
→ AI-Driven Intelligent storage.
→ Multi-Cloud & Data Mobility.
→ Sustainable & Green storage.

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Data Engineering Storage Abstractions

* These are data orginizations & query patterns.
* These are built on top of the data storage systems.
* This is a conceptual layer that hides the complexity of physical storage. Support pipelines, analytics & ML workloads.
* Instead of an application directly dealing with each physical storage device, a storage abstraction layer provides a common interface.

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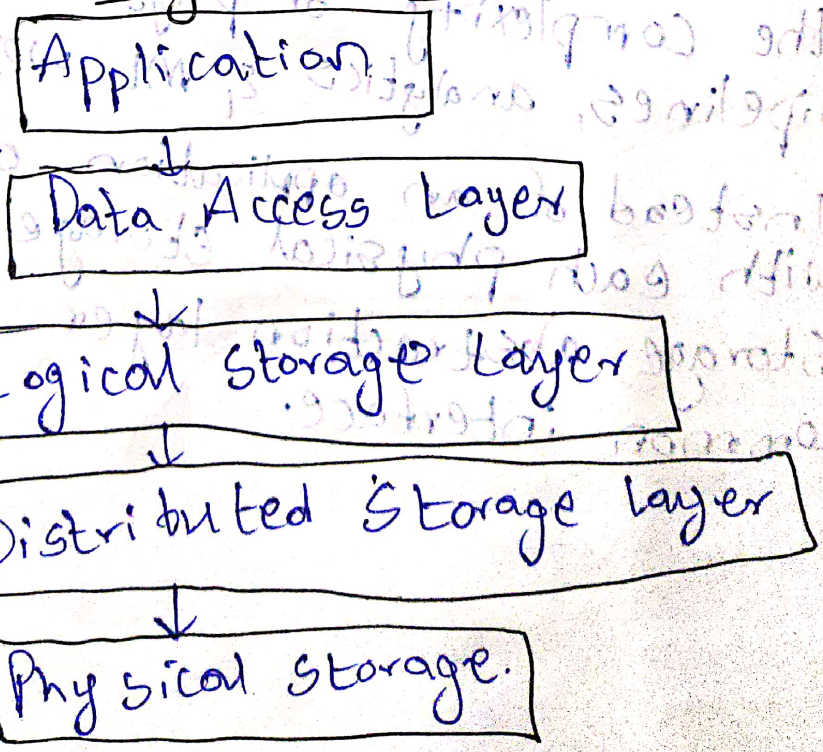
Ex. HDD/SSD Ex. Cloud Storage Ex. MySQL, Oracle.

Features

→ Simplicity
→ Scalability.
→ Portability.
→ Fault tolerance.
→ Performance optimization.
→ Security.

Levels of storage Abstraction

* Each layer hides details from the layer above it.



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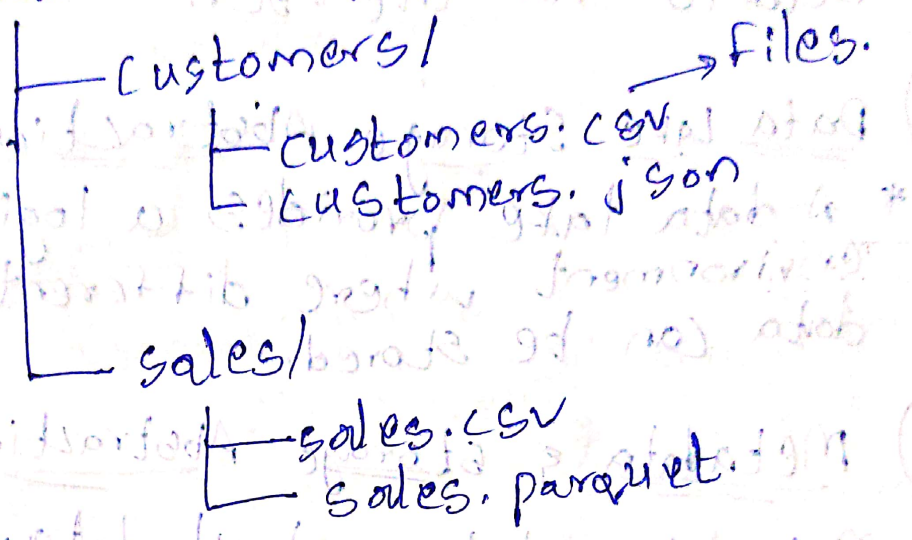
Major Storage Abstractions

~~* In file~~

1) File Storage

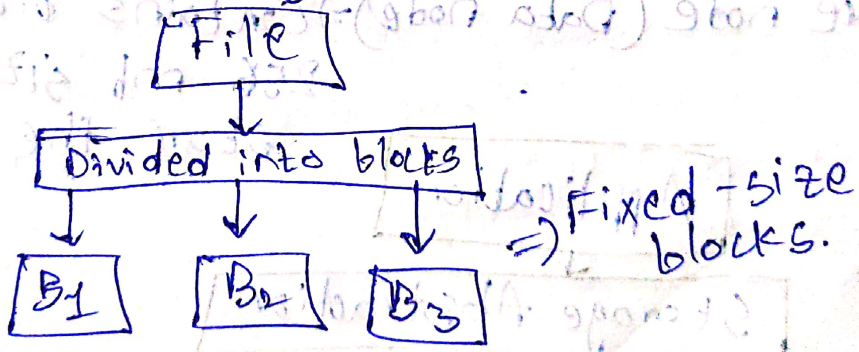
* In file storage, data is organized into files & directories.

Ex:- /data → directory



Ex:- HDFS, NFS & local file systems.

2) Block Storage



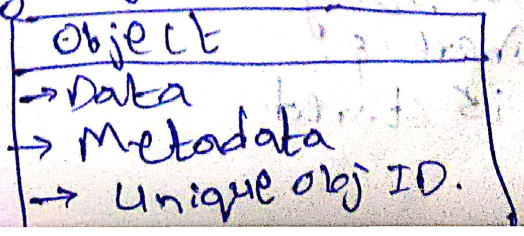
⇒ Fixed-size blocks.

Ex:- SAN, Amazon EBS.

3) Object storage

* Stores data as objects.

→ Generally object contains,



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Ex:- Amazon S3, Azure Blob Storage.

* Object storage can store enormous amounts of data economically.

4.) Database Storage Abstraction

* This hides the physical storage details for the user.

5.) Data Lake Storage Abstraction

* A data lake provides a logical storage environment where different types of data can be stored.

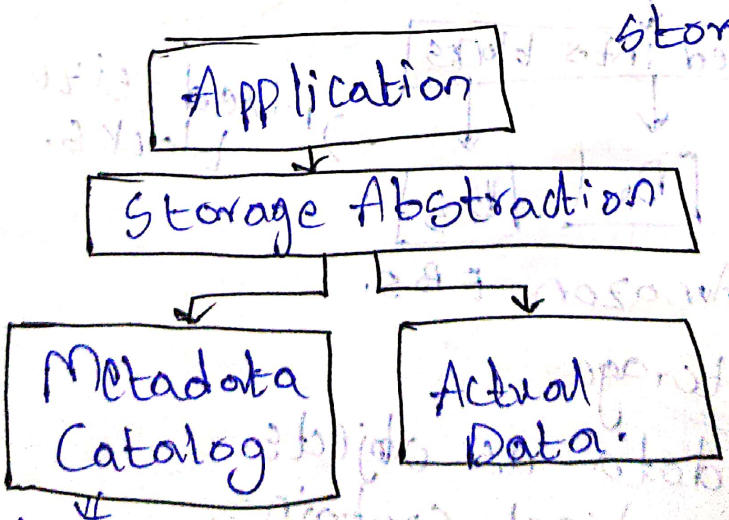
6.) Metadata & Storage Abstraction

Metadata = data about data.

* In HDFS we have two nodes,

1.) Master node (Name node) => Contains metadata

2.) Slave node (Data node) => Contains blocks with 256 mb size to store the data.



This tells processing engines where & how data is stored.

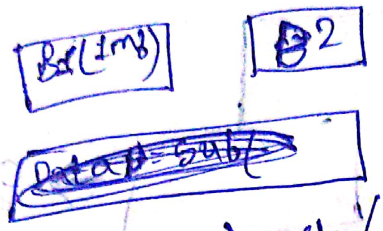
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1) Partitioning & Storage Abstraction

* Large datasets are often divided into partitions.

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Ex:- Data size = 257 Mb need to be stored in big data at data nodes. Contains with number of blocks of default size = 256 mb.



Data D = subset (Data).

* Block B1 is fully occupied with data Do & Block B2 is occupied with 1mb.

→ Remaining free space is B2 = 255 mb.

Note:- Free space is not utilized by any other data.

* Limitations

→ The cost & maintanence is high.

Advantages

→ Retrieval of the data from the blocks is effective.

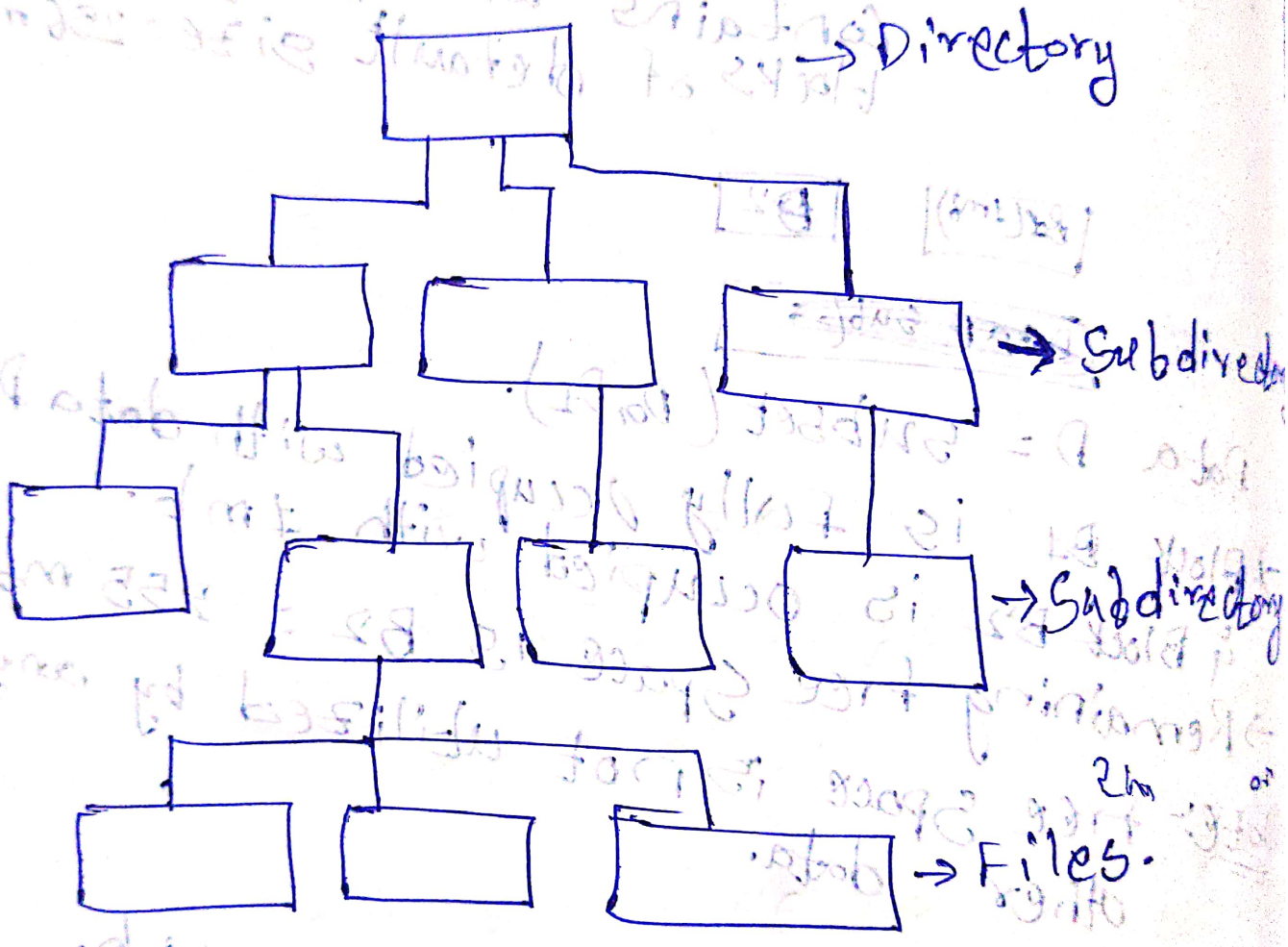
A) 10 26 mb

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* HDFS architecture contains data node, name node, slave node, master node.

3V's ⇒ Volume
Velocity
Variety.

File storage Structure



* Data ware house is a specialized storage system designed primarily for analytics & reporting.

Resilient Distributed Datasets (RDDs)

* This is Spark's fundamental data abstraction - an immutable, partitioned collection of records.

Dataframes

* A distributed collection of data organized into named columns.

Spark SQL

* Lets you run SQL queries directly on DataFrames using the same execution engine.

Usecases

→ Netflix
→ Uber

Apache Flink (RealTime Event Driven Process

* Is an open-source, distributed stream processing framework designed for stateful computations.

Architecture

→ Job Manager
→ Task Managers
→ Dataflow Graph.
→ Parallelism.

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